

1



Setup & Validate Query

canonicalize + validate
input FASTA

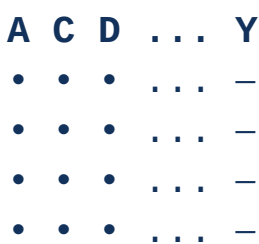
2



Homology Search

jackhmmer search →
raw MSA

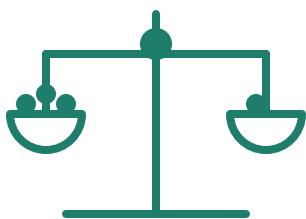
3



Clean & Filter MSA

Stockholm → N×L matrix,
EVEREST filters

4



Sequence Reweighting

99%-identity reweighting,
Neff

5



Build PSSM

per-column amino
acid frequencies

6

pos 1	0.5	-1.2	0.3
pos 2	-0.7	0.8	-1.1
...			
pos N	0.2	-0.4	1.3

Score Variants

log-odds scoring
of DMS variants

7



Evaluate

Spearman ρ vs. DMS assays,
bootstrap CI, scatter